Computational Logic — Solutions & Theory

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1. Theory — Propositional logic: syntax and semantics

Every solution below follows the sheet's format: statement → theory → method → conclusion. This section is the theory you cite in step 2.

1.1 Syntax

- **Propositional variables:** p, q, r, … each takes a truth value T or F.
- **Connectives:** ¬ (not), ∧ (and), ∨ (or), → (implies), ↔ (equivalent). Derived: p ↓ q ≡ ¬(p ∨ q) (NOR), p ↑ q ≡ ¬(p ∧ q) (NAND), p ⊕ q ≡ ¬(p ↔ q) (XOR).
- **Precedence (strongest first):** ¬, ∧, ∨, →, ↔. Example: ¬p ∨ q ∧ r → q ∧ ¬p reads as (¬p ∨ (q ∧ r)) → (q ∧ ¬p).
- → associates to the right: p → q → r means p → (q → r).

1.2 Truth tables of the connectives

p	q	p ∧ q	p ∨ q	p → q	p ↔ q	p ↓ q	p ↑ q	p ⊕ q
T	T	T	T	T	T	F	F	F
T	F	F	T	F	F	F	T	T
F	T	F	T	T	F	F	T	T
F	F	F	F	T	T	T	T	F

Key fact about →: it is false **only** when the premise is T and the conclusion is F. A false premise makes the implication true ("if $x \in \mathbb{N}$ then $x \in \mathbb{Z}$": for $x = -3$, F → T = T).

1.3 Semantic definitions

- **Interpretation** i: {p, q, r, …} → {T, F} — one row of the truth table. With n variables there are 2^n interpretations (rows $i_1 \ldots i_2{}^n$, ordered TTT, TTF, …, FFF).
- **Model** of U: an interpretation with i(U) = T. **Anti-model:** i(U) = F.
- **Tautology (valid formula), ⊨ U:** every interpretation is a model.
- **Consistent (satisfiable):** at least one model.
- **Inconsistent (contradiction):** no model.
- **Contingent:** consistent but not valid (has both models and anti-models).
- **Logical equivalence, U ≡ V:** U and V have identical truth tables (equivalently, ⊨ U ↔ V).

- **Logical consequence, $U_1, \ldots, U_n \models V$:** every model of $U_1 \wedge \ldots \wedge U_n$ is also a model of V. Theorem: $U \models V$ iff $\models U \to V$.

1.4 The truth table method

1. List the variables; write the 2^n rows $i_1, i_2, \ldots$ in the standard order.
2. Add one column per subformula, from the innermost outward (respecting precedence).
3. The last column is the formula. Read the answer from it: all T → tautology; all F → inconsistent; mixed → contingent; T rows are the models, F rows the anti-models.
4. Write at least one model explicitly, e.g. i_2: {p, q, r} → {T, F}, $i_2(p) = T$, $i_2(q) = T$, $i_2(r) = F$.

1.5 Laws of equivalence (used in normal forms and proofs)

Law	Form
Implication	$U \to V \equiv \neg U \vee V$
Equivalence	$U \leftrightarrow V \equiv (U \to V) \wedge (V \to U) \equiv (\neg U \vee V) \wedge (\neg V \vee U)$
Double negation	$\neg\neg U \equiv U$
De Morgan	$\neg(U \wedge V) \equiv \neg U \vee \neg V$; $\neg(U \vee V) \equiv \neg U \wedge \neg V$
Distributivity	$U \wedge (V \vee W) \equiv (U \wedge V) \vee (U \wedge W)$; $U \vee (V \wedge W) \equiv (U \vee V) \wedge (U \vee W)$
Commutativity, associativity	for $\wedge$, $\vee$, $\leftrightarrow$
Excluded middle / contradiction	$U \vee \neg U \equiv T$; $U \wedge \neg U \equiv F$
Absorption	$U \vee (U \wedge V) \equiv U$; $U \wedge (U \vee V) \equiv U$
Contraposition	$U \to V \equiv \neg V \to \neg U$

2. Exercise 1 — Classify the formula (truth table method)

All six formulas are **contingent**: each has both models and anti-models. Theory used: §1.3 definitions and §1.4 method. Watch the precedence: $\neg p \vee q \wedge r$ means $\neg p \vee (q \wedge r)$, and the main connective is always the last →.

$U_1 = q \vee \neg p \vee r \rightarrow \neg(p \vee r)$

i	p	q	r	¬p	q ∨ ¬p	q ∨ ¬p ∨ r	p ∨ r	¬(p ∨ r)	U_1
i1	T	T	T	F	T	T	T	F	F
i2	T	T	F	F	T	T	T	F	F
i3	T	F	T	F	F	T	T	F	F
i4	T	F	F	F	F	F	T	F	T
i5	F	T	T	T	T	T	T	F	F
i6	F	T	F	T	T	T	F	T	T
i7	F	F	T	T	T	T	T	F	F
i8	F	F	F	T	T	T	F	T	T

Conclusion: the last column has both T and F → U_1 is **contingent (consistent, not valid)**.

- Models: i4, i6, i8 (e.g. i4(p) = T, i4(q) = F, i4(r) = F).
- Anti-models: i1, i2, i3, i5, i7.

$U_2 = \neg p \vee \neg(q \wedge r) \rightarrow q \wedge \neg p$

i	p	q	r	¬p	q ∧ r	¬(q ∧ r)	¬p ∨ ¬(q ∧ r)	q ∧ ¬p	U_2
i1	T	T	T	F	T	F	F	F	T
i2	T	T	F	F	F	T	T	F	F
i3	T	F	T	F	F	T	T	F	F
i4	T	F	F	F	F	T	T	F	F
i5	F	T	T	T	T	F	T	T	T
i6	F	T	F	T	F	T	T	T	T
i7	F	F	T	T	F	T	T	F	F
i8	F	F	F	T	F	T	T	F	F

Conclusion: U_2 is **contingent (consistent, not valid)**.

- Models: i1, i5, i6 (e.g. i1(p) = T, i1(q) = T, i1(r) = T).
- Anti-models: i2, i3, i4, i7, i8.

$U_3 = \neg p \wedge (\neg q \vee r) \rightarrow q \wedge \neg p \vee r$

i	p	q	r	¬p	¬q	¬q ∨ r	¬p ∧ (¬q ∨ r)	q ∧ ¬p	q ∧ ¬p ∨ r	U_3
i1	T	T	T	F	F	T	F	F	T	T

i	p	q	r	¬p	¬q	¬q ∨ r	¬p ∧ (¬q ∨ r)	q ∧ ¬p	q ∧ ¬p ∨ r	U_3
i2	T	T	F	F	F	F	F	F	F	T
i3	T	F	T	F	T	T	F	F	T	T
i4	T	F	F	F	T	T	F	F	F	T
i5	F	T	T	T	F	T	T	T	T	T
i6	F	T	F	T	F	F	F	T	T	T
i7	F	F	T	T	T	T	T	F	T	T
i8	F	F	F	T	T	T	T	F	F	F

Conclusion: U_3 is **contingent (consistent, not valid)** — a single anti-model.

- Models: i1–i7 (e.g. i1(p) = T, i1(q) = T, i1(r) = T).
- Anti-models: i8 (p = q = r = F).

$U_4 = \neg(\neg p \wedge q) \vee r \rightarrow p \wedge (\neg q \vee r)$

i	p	q	r	¬p	¬p ∧ q	¬(¬p ∧ q)	¬(¬p ∧ q) ∨ r	¬q	¬q ∨ r	p ∧ (¬q ∨ r)	U_4
i1	T	T	T	F	F	T	T	F	T	T	T
i2	T	T	F	F	F	T	T	F	F	F	F
i3	T	F	T	F	F	T	T	T	T	T	T
i4	T	F	F	F	F	T	T	T	T	T	T
i5	F	T	T	T	T	F	T	F	T	F	F
i6	F	T	F	T	T	F	F	F	F	F	T
i7	F	F	T	T	F	T	T	T	T	F	F
i8	F	F	F	T	F	T	T	T	T	F	F

Conclusion: U_4 is **contingent (consistent, not valid)**.

- Models: i1, i3, i4, i6 (e.g. i6(p) = F, i6(q) = T, i6(r) = F — premise false, so implication true).
- Anti-models: i2, i5, i7, i8.

$U_5 = \neg p \vee q \wedge r \rightarrow q \wedge \neg p$ (solved in seminar)

i	p	q	r	¬p	q ∧ r	¬p ∨ q ∧ r	q ∧ ¬p	U_5
i1	T	T	T	F	T	T	F	F
i2	T	T	F	F	F	F	F	T
i3	T	F	T	F	F	F	F	T

i	p	q	r	¬p	q ∧ r	¬p ∨ q ∧ r	q ∧ ¬p	U_5
i4	T	F	F	F	F	F	F	T
i5	F	T	T	T	T	T	T	T
i6	F	T	F	T	F	T	T	T
i7	F	F	T	T	F	T	F	F
i8	F	F	F	T	F	T	F	F

Conclusion: U_5 is **contingent (consistent, not valid)**.

- Models: i2, i3, i4, i5, i6 (e.g. i2(p) = T, i2(q) = T, i2(r) = F).
- Anti-models: i1, i7, i8.

$U_6 = \neg p \vee (\neg q \wedge \neg r) \rightarrow q \wedge \neg p \wedge r$

i	p	q	r	¬p	¬q	¬r	¬q ∧ ¬r	¬p ∨ (¬q ∧ ¬r)	q ∧ ¬p	q ∧ ¬p ∧ r	U_6
i1	T	T	T	F	F	F	F	F	F	F	T
i2	T	T	F	F	F	T	F	F	F	F	T
i3	T	F	T	F	T	F	F	F	F	F	T
i4	T	F	F	F	T	T	T	T	F	F	F
i5	F	T	T	T	F	F	F	T	T	T	T
i6	F	T	F	T	F	T	F	T	T	F	F
i7	F	F	T	T	T	F	F	T	F	F	F
i8	F	F	F	T	T	T	T	T	F	F	F

Conclusion: U_6 is **contingent (consistent, not valid)**.

- Models: i1, i2, i3, i5 (e.g. i5(p) = F, i5(q) = T, i5(r) = T).
- Anti-models: i4, i6, i7, i8.

3. Exercise 2 — Prove the tautologies

All six formulas are tautologies: their last column is T on all 8 rows. Theory: a formula U is a tautology (⊨ U) iff every interpretation is a model of U (§1.3). Shortcut to check your table: an implication A → B can only be F on a row where A = T and B = F — find those rows and confirm none exist.

2.1 Left-distribution of → over ∧: (p → (q ∧ r)) → ((p → q) ∧ (p → r))

i	p	q	r	q ∧ r	p → (q ∧ r)	p → q	p → r	(p → q) ∧ (p → r)	U
i1	T	T	T	T	T	T	T	T	T
i2	T	T	F	F	F	T	F	F	T
i3	T	F	T	F	F	F	T	F	T
i4	T	F	F	F	F	F	F	F	T
i5	F	T	T	T	T	T	T	T	T
i6	F	T	F	F	T	T	T	T	T
i7	F	F	T	F	T	T	T	T	T
i8	F	F	F	F	T	T	T	T	T

Conclusion: all rows are models → ⊨ U, the formula is a tautology.

2.2 Permutation of the premises: (p → (q → r)) → (q → (p → r))

i	p	q	r	q → r	p → (q → r)	p → r	q → (p → r)	U
i1	T	T	T	T	T	T	T	T
i2	T	T	F	F	F	F	F	T
i3	T	F	T	T	T	T	T	T
i4	T	F	F	T	T	F	T	T
i5	F	T	T	T	T	T	T	T
i6	F	T	F	F	T	T	T	T
i7	F	F	T	T	T	T	T	T
i8	F	F	F	T	T	T	T	T

Conclusion: tautology.

2.3 Reunion of the premises: (p → (q → r)) → (p ∧ q → r)

i	p	q	r	q → r	p → (q → r)	p ∧ q	p ∧ q → r	U
i1	T	T	T	T	T	T	T	T
i2	T	T	F	F	F	T	F	T
i3	T	F	T	T	T	F	T	T
i4	T	F	F	T	T	F	T	T
i5	F	T	T	T	T	F	T	T

i	p	q	r	q → r	p → (q → r)	p ∧ q	p ∧ q → r	U
i6	F	T	F	F	T	F	T	T
i7	F	F	T	T	T	F	T	T
i8	F	F	F	T	T	F	T	T

Conclusion: tautology.

2.4 Separation of the premises: (p ∧ q → r) → (p → (q → r))

i	p	q	r	p ∧ q	p ∧ q → r	q → r	p → (q → r)	U
i1	T	T	T	T	T	T	T	T
i2	T	T	F	T	F	F	F	T
i3	T	F	T	F	T	T	T	T
i4	T	F	F	F	T	T	T	T
i5	F	T	T	F	T	T	T	T
i6	F	T	F	F	T	F	T	T
i7	F	F	T	F	T	T	T	T
i8	F	F	F	F	T	T	T	T

Conclusion: tautology. Together with 2.3: p → (q → r) ≡ p ∧ q → r.

2.5 Left-distribution of ∨ over →: p ∨ (q → r) → ((p ∨ q) → (p ∨ r)) (seminar)

i	p	q	r	q → r	p ∨ (q → r)	p ∨ q	p ∨ r	(p ∨ q) → (p ∨ r)	U
i1	T	T	T	T	T	T	T	T	T
i2	T	T	F	F	T	T	T	T	T
i3	T	F	T	T	T	T	T	T	T
i4	T	F	F	T	T	T	T	T	T
i5	F	T	T	T	T	T	T	T	T
i6	F	T	F	F	F	T	F	F	T
i7	F	F	T	T	T	F	T	T	T
i8	F	F	F	T	T	F	F	T	T

Conclusion: tautology.

2.6 Left-distribution of → over ∨: (p → (q ∨ r)) → ((p → q) ∨ (p → r))

i	p	q	r	q ∨ r	p → (q ∨ r)	p → q	p → r	(p → q) ∨ (p → r)	U
i1	T	T	T	T	T	T	T	T	T
i2	T	T	F	T	T	T	F	T	T
i3	T	F	T	T	T	F	T	T	T
i4	T	F	F	F	F	F	F	F	T
i5	F	T	T	T	T	T	T	T	T
i6	F	T	F	T	T	T	T	T	T
i7	F	F	T	T	T	T	T	T	T
i8	F	F	F	F	T	T	T	T	T

Conclusion: tautology.

4. Exercise 3 — Properties of ↓, ↑, ⊕

Only ⊕ is associative, and both De Morgan laws hold; the other three properties fail. Theory: U ≡ V iff U and V have identical truth tables (§1.3); one row where they differ is enough to disprove the property. Definitions: p ↓ q ≡ ¬(p ∨ q), p ↑ q ≡ ¬(p ∧ q), p ⊕ q ≡ ¬(p ↔ q).

Property	Holds?	Rows where U ≠ V
Associativity of ↑	No	i2, i4, i5, i7
Associativity of ↓	No	i2, i4, i5, i7
Associativity of ⊕	**Yes**	—
Distribution of ↑ over ↓	No	i2, i3, i5, i6, i7, i8
Distribution of ↓ over ↑	No	i1, i2, i3, i4, i6, i7
De Morgan for ↓ and ↑	**Yes** (both)	—

3.1 Associativity of ↑: p ↑ (q ↑ r) ≡ (p ↑ q) ↑ r ?

i	p	q	r	q ↑ r	U = p ↑ (q ↑ r)	p ↑ q	V = (p ↑ q) ↑ r
i1	T	T	T	F	T	F	T
i2	T	T	F	T	F	F	T
i3	T	F	T	T	F	T	F

i	p	q	r	q ↑ r	U = p ↑ (q ↑ r)	p ↑ q	V = (p ↑ q) ↑ r
i4	T	F	F	T	F	T	T
i5	F	T	T	F	T	T	F
i6	F	T	F	T	T	T	T
i7	F	F	T	T	T	T	F
i8	F	F	F	T	T	T	T

Conclusion: the columns differ (e.g. on i2: U = F, V = T) → U ≢ V, ↑ is **not** associative.

3.2 Associativity of ↓: p ↓ (q ↓ r) ≡ (p ↓ q) ↓ r ?

i	p	q	r	q ↓ r	U = p ↓ (q ↓ r)	p ↓ q	V = (p ↓ q) ↓ r
i1	T	T	T	F	F	F	F
i2	T	T	F	F	F	F	T
i3	T	F	T	F	F	F	F
i4	T	F	F	T	F	F	T
i5	F	T	T	F	T	F	F
i6	F	T	F	F	T	F	T
i7	F	F	T	F	T	T	F
i8	F	F	F	T	F	T	F

Conclusion: differ on i2 → ↓ is **not** associative.

3.3 Associativity of ⊕: p ⊕ (q ⊕ r) ≡ (p ⊕ q) ⊕ r ? (seminar)

i	p	q	r	q ⊕ r	U = p ⊕ (q ⊕ r)	p ⊕ q	V = (p ⊕ q) ⊕ r
i1	T	T	T	F	T	F	T
i2	T	T	F	T	F	F	F
i3	T	F	T	T	F	T	F
i4	T	F	F	F	T	T	T
i5	F	T	T	F	F	T	F
i6	F	T	F	T	T	T	T
i7	F	F	T	T	T	F	T
i8	F	F	F	F	F	F	F

Conclusion: identical columns → U ≡ V, ⊕ **is** associative. (Intuition: p ⊕ q ⊕ r is T exactly when an odd number of variables are T.)

3.4 Distribution of ↑ over ↓: $p \uparrow (q \downarrow r) \equiv (p \uparrow q) \downarrow (p \uparrow r)$?

i	p	q	r	$q \downarrow r$	$U = p \uparrow (q \downarrow r)$	$p \uparrow q$	$p \uparrow r$	$V = (p \uparrow q) \downarrow (p \uparrow r)$
i1	T	T	T	F	T	F	F	T
i2	T	T	F	F	T	F	T	F
i3	T	F	T	F	T	T	F	F
i4	T	F	F	T	F	T	T	F
i5	F	T	T	F	T	T	T	F
i6	F	T	F	F	T	T	T	F
i7	F	F	T	F	T	T	T	F
i8	F	F	F	T	T	T	T	F

Conclusion: differ on i2 → the distribution does **not** hold.

3.5 Distribution of ↓ over ↑: $p \downarrow (q \uparrow r) \equiv (p \downarrow q) \uparrow (p \downarrow r)$? (seminar)

i	p	q	r	$q \uparrow r$	$U = p \downarrow (q \uparrow r)$	$p \downarrow q$	$p \downarrow r$	$V = (p \downarrow q) \uparrow (p \downarrow r)$
i1	T	T	T	F	F	F	F	T
i2	T	T	F	T	F	F	F	T
i3	T	F	T	T	F	F	F	T
i4	T	F	F	T	F	F	F	T
i5	F	T	T	F	T	F	F	T
i6	F	T	F	T	F	F	T	T
i7	F	F	T	T	F	T	F	T
i8	F	F	F	T	F	T	T	F

Conclusion: differ on i1 (U = F, V = T) → does **not** hold.

3.6 De Morgan's laws for ↓ and ↑

i	p	q	$p \downarrow q$	$\neg(p \downarrow q)$	$\neg p$	$\neg q$	$\neg p \uparrow \neg q$	$p \uparrow q$	$\neg(p \uparrow q)$	$\neg p \downarrow \neg q$
i1	T	T	F	T	F	F	T	F	T	T
i2	T	F	F	T	F	T	T	T	F	F
i3	F	T	F	T	T	F	T	T	F	F
i4	F	F	T	F	T	T	F	T	F	F

Conclusion: columns ¬(p ↓ q) and ¬p ↑ ¬q are identical, and so are ¬(p ↑ q) and ¬p ↓ ¬q → both laws **hold**. (Both sides of the first equal p ∨ q; both sides of the second equal p ∧ q.)

5. Exercise 4 — Logical consequence (truth table method)

All six consequences hold. Theory: U ⊨ V (V is a logical consequence of U) iff every model of U is also a model of V, i.e. ∀ i: i(U) = T ⇒ i(V) = T. Method: build U and V in one table, mark the rows where U = T, and check V = T on each of them. Rows where U = F don't matter. Equivalent test: U ⊨ V iff ⊨ U → V.

4.1 $p \to q \models (p \to r) \to (p \to q \wedge r)$

i	p	q	r	U = p→q	p→r	q∧r	p→q∧r	V
i1	T	T	T	**T**	T	T	T	**T**
i2	T	T	F	**T**	F	F	F	**T**
i3	T	F	T	F	T	F	F	F
i4	T	F	F	F	F	F	F	T
i5	F	T	T	**T**	T	T	T	**T**
i6	F	T	F	**T**	T	F	T	**T**
i7	F	F	T	**T**	T	F	T	**T**
i8	F	F	F	**T**	T	F	T	**T**

Conclusion: models of U are i1, i2, i5–i8, and V = T on all of them → U ⊨ V holds.

4.2 $p \to q \models (q \to r) \to (p \to r)$

i	p	q	r	U = p→q	q→r	p→r	V
i1	T	T	T	**T**	T	T	**T**
i2	T	T	F	**T**	F	F	**T**
i3	T	F	T	F	T	T	T
i4	T	F	F	F	T	F	F
i5	F	T	T	**T**	T	T	**T**
i6	F	T	F	**T**	F	T	**T**
i7	F	F	T	**T**	T	T	**T**
i8	F	F	F	**T**	T	T	**T**

Conclusion: holds (this is the transitivity / hypothetical syllogism).

4.3 $p \to (q \to r) \models (p \to q) \to (p \to r)$

i	p	q	r	q → r	U = p → (q → r)	p → q	p → r	V
i1	T	T	T	T	**T**	T	T	**T**
i2	T	T	F	F	F	T	F	F
i3	T	F	T	T	**T**	F	T	**T**
i4	T	F	F	T	**T**	F	F	**T**
i5	F	T	T	T	**T**	T	T	**T**
i6	F	T	F	F	**T**	T	T	**T**
i7	F	F	T	T	**T**	T	T	**T**
i8	F	F	F	T	**T**	T	T	**T**

Conclusion: holds (the self-distribution of →, axiom A2 of the deductive system).

4.4 $p \to r \models (q \to r) \to ((p \vee q) \to r)$

i	p	q	r	U = p → r	q → r	p ∨ q	p ∨ q → r	V
i1	T	T	T	**T**	T	T	T	**T**
i2	T	T	F	F	F	T	F	T
i3	T	F	T	**T**	T	T	T	**T**
i4	T	F	F	F	T	T	F	F
i5	F	T	T	**T**	T	T	T	**T**
i6	F	T	F	**T**	F	T	F	**T**
i7	F	F	T	**T**	T	F	T	**T**
i8	F	F	F	**T**	T	F	T	**T**

Conclusion: holds (proof by cases).

4.5 $p \to q \models (q \to r) \to (p \to q \vee r)$ (seminar)

i	p	q	r	U = p → q	q → r	q ∨ r	p → q ∨ r	V
i1	T	T	T	**T**	T	T	T	**T**
i2	T	T	F	**T**	F	T	T	**T**
i3	T	F	T	F	T	T	T	T
i4	T	F	F	F	T	F	F	F
i5	F	T	T	**T**	T	T	T	**T**

i	p	q	r	U = p → q	q → r	q ∨ r	p → q ∨ r	V
i6	F	T	F	**T**	F	T	T	**T**
i7	F	F	T	**T**	T	T	T	**T**
i8	F	F	F	**T**	T	F	T	**T**

Conclusion: holds. Note: V alone is not a tautology (i4), so we really need U — the anti-model i4 of V is not a model of U.

4.6 $r \to (q \to p) \models (r \to q) \to (r \to p)$

i	p	q	r	q → p	U = r → (q → p)	r → q	r → p	V
i1	T	T	T	T	**T**	T	T	**T**
i2	T	T	F	T	**T**	T	T	**T**
i3	T	F	T	T	**T**	F	T	**T**
i4	T	F	F	T	**T**	T	T	**T**
i5	F	T	T	F	F	T	F	F
i6	F	T	F	F	**T**	T	T	**T**
i7	F	F	T	T	**T**	F	F	**T**
i8	F	F	F	T	**T**	T	T	**T**

Conclusion: holds (same pattern as 4.3 with r as the common premise).

6. Exercise 5 — From natural language to formulas

Each sentence becomes one formula; its models follow from the truth table of the main connective. Translation rules:

Natural language	Formula	Why
if A then B · B if A · provided that A, B	A → B	A is the sufficient condition
A only if B	A → B (≡ ¬B → ¬A)	B is the necessary condition
A if and only if B	A ↔ B	both directions
A and B · A but B	A ∧ B	
A or B · either A or B (inclusive)	A ∨ B	
not A	¬A	

Shortcut for models: A → B has exactly the anti-models where A = T and B = F; A ↔ B has as models exactly the rows where A and B agree. Variables are listed in the order shown; a triple (T, F, T) gives their values.

Set 1 (seminar)

S1: If it is sunny and warm then I will go swimming. s: "it is sunny", w: "it is warm", m: "I will go swimming". **S1 = s ∧ w → m**.

- Anti-model (s, w, m) = (T, T, F) — the only one. Models: the other 7 rows.

S2: Mary and Jane will go for a walk only if it is sunny and not windy. k: "Mary and Jane go for a walk", s: "it is sunny", w: "it is windy". **S2 = k → s ∧ ¬w**.

- Anti-models (k, s, w): (T, T, T), (T, F, T), (T, F, F). Models: (T, T, F) and all 4 rows with k = F.

S3: John goes to the cinema iff it is Saturday and it is raining. j, s, r. **S3 = j ↔ s ∧ r**.

- Models (j, s, r): (T, T, T), (F, T, F), (F, F, T), (F, F, F). Anti-models: (T, T, F), (T, F, T), (T, F, F), (F, T, T).

Set 2

S1 = k ↔ c ∧ o — k: "we go skating", c: "it is cold outside", o: "the rink is open". Models (k, c, o): (T, T, T), (F, T, F), (F, F, T), (F, F, F); the other 4 are anti-models.

S2 = e ∧ l → z — e: "you have experience", l: "you like the cold", z: "skating is easy". Only anti-model (e, l, z) = (T, T, F).

S3 = n → h ∨ b — n: "you can enter the rink", h: "you bring skates from home", b: "you borrow skates". Only anti-model (n, h, b) = (T, F, F).

Set 3 (seminar: S1, S2)

S1 = d ∧ c → m — d: "Sherlock notices small details", c: "he catches the suspect", m: "he solves the mystery". "m if A" means A → m. Only anti-model (d, c, m) = (T, T, F).

S2 = l → g — l: "Sherlock listens to Watson", g: "Watson has good ideas" (equivalently ¬g → ¬l, contraposition). Only anti-model (l, g) = (T, F); models (T, T), (F, T), (F, F).

S3 = w ↔ t ∨ n — w: "Watson intervenes", t: "Sherlock is in trouble", n: "Watson has a new insight". Models (w, t, n): (T, T, T), (T, T, F), (T, F, T), (F, F, F). Anti-models: (T, F, F), (F, T, T), (F, T, F), (F, F, T).

Set 4

S1 = f ↔ c ∧ t — f: "Mary starts a fantasy book", c: "she finishes her current book", t: "she has free time". Models (f, c, t): (T, T, T), (F, T, F), (F, F, T), (F, F, F).

S2 = g ∨ b → j — g: "the book has good reviews", b: "it is a bestseller", j: "Jane reads the book". Anti-models (g, b, j): (T, T, F), (T, F, F), (F, T, F); the other 5 rows are models.

S3 = m → s ∧ o — m: "Charlotte reads mystery books", s: "it is Sunday", o: "she is bored". Anti-models (m, s, o): (T, T, F), (T, F, T), (T, F, F); models: (T, T, T) + the 4 rows with m = F.

Set 5

S1 = h → o ∧ m — h: "the wizard wears his starry hat", o: "he is outside", m: "he is practicing magic". Anti-models (h, o, m): (T, T, F), (T, F, T), (T, F, F).

S2 = z ↔ e ∨ u — z: "the beard sparkles", e: "he drinks enchanted tea", u: "a unicorn sneezes nearby". Models (z, e, u): (T, T, T), (T, T, F), (T, F, T), (F, F, F).

S3 = i ∨ y → ¬s — i: "wrong ingredients", y: "says the wrong words", s: "the spell works". Anti-models (i, y, s): (T, T, T), (T, F, T), (F, T, T); the other 5 are models.

Set 6

S1 = l ∨ j → c — l: "the team is losing", j: "a player gets injured", c: "the coach changes the strategy". Anti-models (l, j, c): (T, T, F), (T, F, F), (F, T, F).

S2 = m → d ∧ s — m: "the team makes a comeback", d: "it improves its defense", s: "it scores more". Anti-models (m, d, s): (T, T, F), (T, F, T), (T, F, F).

S3 = a ↔ w — a: "the team advances", w: "it wins the next game by 2 or more points". Models (a, w): (T, T), (F, F); anti-models: (T, F), (F, T).

7. Theory — Normal forms

A normal form turns the truth-table question into a syntactic check: CNF answers "valid?" and gives anti-models, DNF answers "inconsistent?" and gives models.

7.1 Definitions

- **Literal:** a variable or its negation (p, ¬p).
- **Clause:** a disjunction of literals (p ∨ ¬q ∨ r). **Cube:** a conjunction of literals (p ∧ ¬q ∧ r).
- **CNF (conjunctive normal form):** a conjunction of clauses, $C_1 \wedge C_2 \wedge \dots \wedge C_n$.
- **DNF (disjunctive normal form):** a disjunction of cubes, $K_1 \vee K_2 \vee \dots \vee K_n$.
- Every formula has an equivalent CNF and an equivalent DNF.

7.2 Normalization algorithm

1. **Eliminate → and ↔:** U → V ≡ ¬U ∨ V; U ↔ V ≡ (¬U ∨ V) ∧ (¬V ∨ U).

2. **Push negations inward:** De Morgan ¬(U ∧ V) ≡ ¬U ∨ ¬V, ¬(U ∨ V) ≡ ¬U ∧ ¬V, and ¬¬U ≡ U, until ¬ sits only on variables.
3. **Distribute:** for DNF use U ∧ (V ∨ W) ≡ (U ∧ V) ∨ (U ∧ W); for CNF use U ∨ (V ∧ W) ≡ (U ∨ V) ∧ (U ∨ W).
4. Optionally simplify: drop repeated literals, absorption (U ∧ (U ∨ V) ≡ U), a clause with p ∨ ¬p is always T and can be dropped from a CNF; a cube with p ∧ ¬p is always F and can be dropped from a DNF.

7.3 The four theorems used in Ex 6–9

Question	Normal form	Criterion
Is U **valid**? (Ex 6)	CNF	U is valid iff **every clause** contains a pair of opposite literals (p ∨ ¬p)
Is U **inconsistent**? (Ex 8)	DNF	U is inconsistent iff **every cube** contains a pair of opposite literals (p ∧ ¬p)
Models of U (Ex 7)	DNF	each consistent cube is T exactly on the interpretations that make all its literals T; models = union over cubes
Anti-models of U (Ex 9)	CNF	each clause is F exactly on the interpretations that make all its literals F; anti-models = union over clauses

Row order used everywhere: i1 = TTT, i2 = TTF, i3 = TFT, i4 = TFF, i5 = FTT, i6 = FTF, i7 = FFT, i8 = FFF for (p, q, r). Example: the cube p ∧ ¬r fixes p = T, r = F, q free → models i2, i4.

8. Exercise 6 — Validity via CNF

All six formulas are valid: in each CNF every clause contains a complementary pair. Steps: (1) eliminate →, (2) De Morgan, (3) distribute. The DNF appears naturally after step 2; the CNF comes from distributing ∨ over ∧.

$U_1 = (p \to \neg q) \wedge (q \vee r) \to (p \to r)$

- $U_1 \equiv \neg[(\neg p \vee \neg q) \wedge (q \vee r)] \vee (\neg p \vee r) \equiv (p \wedge q) \vee (\neg q \wedge \neg r) \vee \neg p \vee r$ = **DNF** (4 cubes).
- Distribute (p ∧ q) ∨ (¬q ∧ ¬r) ≡ (p ∨ ¬q) ∧ (p ∨ ¬r) ∧ (q ∨ ¬q) ∧ (q ∨ ¬r), then add ¬p ∨ r to each clause:
- **CNF** = (p ∨ ¬q ∨ ¬p ∨ r) ∧ (p ∨ ¬r ∨ ¬p ∨ r) ∧ (q ∨ ¬q ∨ ¬p ∨ r) ∧ (q ∨ ¬r ∨ ¬p ∨ r).
- **Conclusion:** pairs p/¬p, p/¬p, q/¬q, r/¬r in the four clauses → $\models U_1$.

$U_2 = (p \to q) \land (p \land q \to r) \to (p \to r)$

- $U_2 \equiv \neg(\neg p \lor q) \lor \neg(\neg p \lor \neg q \lor r) \lor \neg p \lor r \equiv (p \land \neg q) \lor (p \land q \land \neg r) \lor \neg p \lor r$ = **DNF**.
- **CNF** = $(p \lor \neg p \lor r) \land (p \lor q \lor \neg p \lor r) \land (p \lor \neg r \lor \neg p \lor r) \land (\neg q \lor p \lor \neg p \lor r) \land (\neg q \lor q \lor \neg p \lor r) \land (\neg q \lor \neg r \lor \neg p \lor r)$.
- **Conclusion:** every clause has p/¬p, q/¬q or r/¬r → $\models U_2$.

$U_3 = (p \land q \to r) \to (p \to (q \to r))$

- $U_3 \equiv \neg(\neg p \lor \neg q \lor r) \lor \neg p \lor \neg q \lor r \equiv (p \land q \land \neg r) \lor \neg p \lor \neg q \lor r$ = **DNF**.
- **CNF** = $(p \lor \neg p \lor \neg q \lor r) \land (q \lor \neg p \lor \neg q \lor r) \land (\neg r \lor \neg p \lor \neg q \lor r)$.
- **Conclusion:** pairs p/¬p, q/¬q, r/¬r → $\models U_3$. (Pattern: U = ¬A ∨ A with A = ¬p ∨ ¬q ∨ r.)

$U_4 = (p \to (q \lor r)) \to ((p \to q) \lor (p \to r))$

- $U_4 \equiv \neg(\neg p \lor q \lor r) \lor \neg p \lor q \lor \neg p \lor r \equiv (p \land \neg q \land \neg r) \lor \neg p \lor q \lor r$ = **DNF**.
- **CNF** = $(p \lor \neg p \lor q \lor r) \land (\neg q \lor \neg p \lor q \lor r) \land (\neg r \lor \neg p \lor q \lor r)$.
- **Conclusion:** $\models U_4$.

$U_7 = (p \to (q \land r)) \to ((p \to q) \land (p \to r))$ (seminar)

- $U_7 \equiv \neg(\neg p \lor (q \land r)) \lor ((\neg p \lor q) \land (\neg p \lor r)) \equiv p \land (\neg q \lor \neg r) \lor ((\neg p \lor q) \land (\neg p \lor r))$.
- **DNF** = $(p \land \neg q) \lor (p \land \neg r) \lor \neg p \lor (q \land r)$. *Correction to the seminar notes: the cubes are $p \land \neg r$ and $q \land r$, not disjunctions — $(\neg p \lor q) \land (\neg p \lor r) \equiv \neg p \lor (q \land r)$.*
- **CNF** = $(p \lor \neg p \lor q) \land (\neg q \lor \neg r \lor \neg p \lor q) \land (p \lor \neg p \lor r) \land (\neg q \lor \neg r \lor \neg p \lor r)$.
- **Conclusion:** each clause has a pair → $\models U_7$.

$U_8 = p \lor (q \to r) \to ((p \lor q) \to (p \lor r))$

- $U_8 \equiv \neg(p \lor \neg q \lor r) \lor \neg(p \lor q) \lor p \lor r \equiv (\neg p \land q \land \neg r) \lor (\neg p \land \neg q) \lor p \lor r$ = **DNF**.
- Distribute $(\neg p \land q \land \neg r) \lor (\neg p \land \neg q) \equiv \neg p \land (\neg p \lor \neg q) \land (q \lor \neg p) \land (q \lor \neg q) \land (\neg r \lor \neg p) \land (\neg r \lor \neg q)$, then add p ∨ r:
- **CNF** = $(\neg p \lor p \lor r) \land (\neg p \lor \neg q \lor p \lor r) \land (q \lor \neg p \lor p \lor r) \land (q \lor \neg q \lor p \lor r) \land (\neg r \lor \neg p \lor p \lor r) \land (\neg r \lor \neg q \lor p \lor r)$.
- **Conclusion:** $\models U_8$.

9. Exercise 7 — Models via DNF

Method: bring U to DNF; each cube gives the rows where all its literals are true; the models are the union.

Formula	DNF	Models
$U_1 = (p \vee q \to r) \to (p \to r) \wedge q$	$(p \wedge \neg r) \vee (q \wedge \neg r) \vee (\neg p \wedge q) \vee (q \wedge r)$	i1, i2, i4, i5, i6
$U_2 = \neg(\neg p \vee q) \vee r \to \neg p \wedge \neg(q \wedge r)$	$(\neg p \wedge \neg r) \vee (q \wedge \neg r) \vee (\neg p \wedge \neg q)$	i2, i6, i7, i8
$U_3 = (p \wedge q \to r) \to (p \to r) \wedge q$	$(p \wedge q \wedge \neg r) \vee (\neg p \wedge q) \vee (q \wedge r)$	i1, i2, i5, i6
$U_4 = (p \vee q) \wedge \neg r \to p \wedge q \wedge r$	$(\neg p \wedge \neg q) \vee r \vee (p \wedge q \wedge r)$	i1, i3, i5, i7, i8
$U_5 = p \vee \neg(q \wedge \neg r) \to p \wedge q \wedge \neg r$	$(\neg p \wedge q \wedge \neg r) \vee (p \wedge q \wedge \neg r)$	i2, i6
$U_6 = (p \vee q \to r) \to (q \to r) \wedge p$	$(p \wedge \neg r) \vee (q \wedge \neg r) \vee (p \wedge \neg q) \vee (p \wedge r)$	i1, i2, i3, i4, i6

Worked derivation for U_1:

1. Eliminate →: $U_1 \equiv \neg(\neg(p \vee q) \vee r) \vee ((\neg p \vee r) \wedge q)$.
2. De Morgan: $\neg(\neg(p \vee q) \vee r) \equiv (p \vee q) \wedge \neg r$.
3. Distribute: $(p \vee q) \wedge \neg r \equiv (p \wedge \neg r) \vee (q \wedge \neg r)$; $(\neg p \vee r) \wedge q \equiv (\neg p \wedge q) \vee (r \wedge q)$.
4. DNF = $(p \wedge \neg r) \vee (q \wedge \neg r) \vee (\neg p \wedge q) \vee (q \wedge r)$.
5. Cubes → rows: $p \wedge \neg r$ → i2, i4; $q \wedge \neg r$ → i2, i6; $\neg p \wedge q$ → i5, i6; $q \wedge r$ → i1, i5.
6. **Conclusion:** models of U_1 = {i1, i2, i4, i5, i6}; e.g. i4(p) = T, i4(q) = F, i4(r) = F.

The other rows of the table follow the same four steps. Key De Morgan moves: U_2 — $\neg((p \wedge \neg q) \vee r) \equiv (\neg p \vee q) \wedge \neg r$; U_4 — $\neg((p \vee q) \wedge \neg r) \equiv (\neg p \wedge \neg q) \vee r$; U_5 — $\neg(p \vee \neg q \vee r) \equiv \neg p \wedge q \wedge \neg r$, so $U_5 \equiv q \wedge \neg r$.

10. Exercise 8 — Inconsistency via DNF

All six formulas are inconsistent: every cube of their DNF contains a complementary pair. Each one is a conjunction of hypotheses with the negation of a valid consequence, so it can never be true.

$U_1 = (p \to (q \to r)) \wedge \neg((p \to q) \to (p \to r))$

- $U_1 \equiv (\neg p \vee \neg q \vee r) \wedge (\neg p \vee q) \wedge \neg(\neg p \vee r) \equiv (\neg p \vee \neg q \vee r) \wedge (\neg p \vee q) \wedge p \wedge \neg r$.
- Distribute (3 × 2 = 6 cubes, each $\wedge\, p \wedge \neg r$): $(\neg p \wedge p \wedge \neg r) \vee (\neg p \wedge q \wedge p \wedge \neg r) \vee (\neg q \wedge \neg p \wedge p \wedge \neg r) \vee (\neg q \wedge q \wedge p \wedge \neg r) \vee (r \wedge \neg p \wedge p \wedge \neg r) \vee (r \wedge q \wedge p \wedge \neg r)$.
- **Conclusion:** cubes contain $p/\neg p$, $p/\neg p$, $p/\neg p$, $q/\neg q$, $r/\neg r$, $r/\neg r$ → U_1 is inconsistent.

$U_3 = (p \to q) \wedge (p \wedge q \to r) \wedge (p \wedge \neg r)$

- $U_3 \equiv (\neg p \vee q) \wedge (\neg p \vee \neg q \vee r) \wedge p \wedge \neg r$.

- DNF: (¬p ∧ p ∧ ¬r) ∨ (¬p ∧ ¬q ∧ p ∧ ¬r) ∨ (¬p ∧ r ∧ p ∧ ¬r) ∨ (q ∧ ¬p ∧ p ∧ ¬r) ∨ (q ∧ ¬q ∧ p ∧ ¬r) ∨ (q ∧ r ∧ p ∧ ¬r).
- **Conclusion:** every cube has p/¬p, q/¬q or r/¬r → inconsistent.

$U_4 = (p \to (q \vee r)) \wedge (\neg(p \to q) \wedge \neg(p \to r))$

- U_4 ≡ (¬p ∨ q ∨ r) ∧ p ∧ ¬q ∧ ¬r.
- DNF: (¬p ∧ p ∧ ¬q ∧ ¬r) ∨ (q ∧ p ∧ ¬q ∧ ¬r) ∨ (r ∧ p ∧ ¬q ∧ ¬r).
- **Conclusion:** inconsistent.

$U_5 = p \wedge (q \to r) \wedge ((p \wedge q) \wedge \neg(p \wedge r))$

- U_5 ≡ p ∧ (¬q ∨ r) ∧ q ∧ (¬p ∨ ¬r).
- DNF: (p ∧ ¬q ∧ q ∧ ¬p) ∨ (p ∧ ¬q ∧ q ∧ ¬r) ∨ (p ∧ r ∧ q ∧ ¬p) ∨ (p ∧ r ∧ q ∧ ¬r).
- **Conclusion:** inconsistent.

$U_6 = (p \to (q \to r)) \wedge (p \wedge q \wedge \neg r)$

- U_6 ≡ (¬p ∨ ¬q ∨ r) ∧ p ∧ q ∧ ¬r.
- DNF: (¬p ∧ p ∧ q ∧ ¬r) ∨ (¬q ∧ p ∧ q ∧ ¬r) ∨ (r ∧ p ∧ q ∧ ¬r).
- **Conclusion:** inconsistent.

$U_7 = (p \to (q \to r)) \wedge \neg(q \to (p \to r))$

- ¬(q → (p → r)) ≡ ¬(¬q ∨ ¬p ∨ r) ≡ q ∧ p ∧ ¬r, so U_7 ≡ (¬p ∨ ¬q ∨ r) ∧ p ∧ q ∧ ¬r — the same as U_6.
- **Conclusion:** inconsistent (same three cubes as U_6).

11. Exercise 9 — Anti-models via CNF

Method: bring U to CNF (drop clauses with p ∨ ¬p — they are never false); each remaining clause is F exactly when all its literals are F; the anti-models are the union. U_5 is valid, so it has none.

Formula	CNF (simplified)	Anti-models
U_1 = (q ∧ r → p) → (p → r) ∧ q	(¬p ∨ q ∨ r) ∧ (¬p ∨ r) ∧ q ∧ (q ∨ r) ∧ (¬p ∨ q) ≡ (¬p ∨ r) ∧ q	i2, i3, i4, i7, i8
U_2 = (q ∨ r → p) → (p → r) ∧ q	(¬p ∨ q ∨ r) ∧ (q ∨ r) ∧ (¬p ∨ r) ∧ (¬p ∨ q)	i2, i3, i4, i8

Formula	CNF (simplified)	Anti-models
$U_3 = (p \vee q \to r) \to (q \to r) \wedge p$	$(p \vee q) \wedge (p \vee \neg r)$	i5, i7, i8
$U_4 = p \vee \neg(q \wedge \neg r) \to p \wedge q \wedge \neg r$	$q \wedge \neg r$	i1, i3, i4, i5, i7, i8
$U_5 = p \wedge \neg(q \wedge \neg r) \to p \vee q \wedge \neg r$	$(\neg p \vee p \vee q) \wedge (\neg p \vee p \vee \neg r)$ — valid	none
$U_6 = (p \wedge q \to r) \to (p \to r) \wedge q$	$(p \vee q) \wedge (\neg p \vee q \vee r) \wedge q \wedge (q \vee \neg r) \equiv q$	i3, i4, i7, i8

Worked derivation for U_2:

1. $U_2 \equiv \neg(\neg(q \vee r) \vee p) \vee ((\neg p \vee r) \wedge q) \equiv ((q \vee r) \wedge \neg p) \vee ((\neg p \vee r) \wedge q)$.
2. Distribute $\vee$ over $\wedge$ (2 × 2 clauses): $(q \vee r \vee \neg p \vee r) \wedge (q \vee r \vee q) \wedge (\neg p \vee \neg p \vee r) \wedge (\neg p \vee q)$.
3. CNF = $(\neg p \vee q \vee r) \wedge (q \vee r) \wedge (\neg p \vee r) \wedge (\neg p \vee q)$.
4. Clauses → falsifying rows: $\neg p \vee q \vee r$ is F for p = T, q = F, r = F → i4; $q \vee r$ → i4, i8; $\neg p \vee r$ → i2, i4; $\neg p \vee q$ → i3, i4.
5. **Conclusion:** anti-models of U_2 = {i2, i3, i4, i8}; e.g. i8(p) = F, i8(q) = F, i8(r) = F.

U_5: $U_5 \equiv \neg p \vee (q \wedge \neg r) \vee p \vee (q \wedge \neg r)$; every clause of its CNF contains $p \vee \neg p$, so U_5 is a tautology and has **no** anti-models.

12. Theory — Deduction (the syntactic method)

A deduction is a finite list of formulas where each line is justified; no truth values are used. Check the exact names and form of axiom A3 against your lecture — the content below is the standard system.

12.1 Definition of deduction

V is deducible from the hypotheses $U_1, \ldots, U_n$ (written $U_1, \ldots, U_n \vdash V$) if there is a sequence $f_1, \ldots, f_m$ with $f_m = V$ and, for every $i = 1, \ldots, m$, one of:

1. f_i is an axiom;
2. $f_i \in \{U_1, \ldots, U_n\}$ (a hypothesis);
3. f_i is obtained from earlier formulas f_j, f_k ($j, k < i$) by an inference rule ($f_j, f_k \vdash f_i$).

With no hypotheses, $\vdash V$ means V is a **theorem**.

12.2 Axioms and rules

- **A1:** U → (V → U)
- **A2:** (U → (V → Z)) → ((U → V) → (U → Z))
- **A3:** (U → V) → (¬V → ¬U)
- **Modus ponens (mp):** U, U → V ⊢ V — the only primitive rule.

Derived rules you may cite (each is provable from A1–A3 + mp):

Rule	Scheme
Modus tollens (mt)	U → V, ¬V ⊢ ¬U
Hypothetical syllogism (sh)	U → V, V → Z ⊢ U → Z
Disjunctive syllogism (ds)	U ∨ V, ¬U ⊢ V
∧-elimination (∧e)	U ∧ V ⊢ U; U ∧ V ⊢ V
∧-introduction (∧i)	U, V ⊢ U ∧ V
∨-introduction (∨i)	U ⊢ U ∨ V
Replacement by equivalents	a subformula may be replaced by an equivalent one (§1.5), e.g. U ∨ V ≡ ¬U → V, U ≡ ¬¬U

12.3 The deduction theorem and its reverse

- **Deduction theorem:** if $U_1, \ldots, U_n, A \vdash B$ then $U_1, \ldots, U_n \vdash A \to B$.
- **Reverse:** if $U_1, \ldots, U_n \vdash A \to B$ then $U_1, \ldots, U_n, A \vdash B$.
- **Method for $\vdash A_1 \to (A_2 \to (\ldots \to (A_k \to B)))$**: move every premise into the hypotheses, prove $A_1, \ldots, A_k \vdash B$, then apply the deduction theorem k times. If the conclusion has another shape (a disjunction), first rewrite it as an implication: X ∨ Y ≡ ¬X → Y.

12.4 Soundness and completeness

$U_1, \ldots, U_n \vdash V$ iff $U_1, \ldots, U_n \models V$. So the consequences proved by truth tables in Ex 4 can also be deduced, and vice versa.

13. Exercise 10 — Deductions by definition

Each deduction below is a sequence $f_1, \ldots, f_m$; the last line is the conclusion. All six hold.

10.1 p → q, r → t, p ∨ r, ¬q ⊢ t

	Formula	Justification
f1	p → q	hyp
f2	¬q	hyp
f3	¬p	mt(f1, f2)
f4	p ∨ r	hyp
f5	r	ds(f4, f3)
f6	r → t	hyp
f7	t	mp(f5, f6)

10.2 p → r, p ∨ r → q, r ⊢ q — the hypothesis p → r is not needed.

	Formula	Justification
f1	r	hyp
f2	p ∨ r	∨i(f1)
f3	p ∨ r → q	hyp
f4	q	mp(f2, f3)

10.3 q → p, t → r, q ∨ t, ¬p ⊢ r

	Formula	Justification
f1	q → p	hyp
f2	¬p	hyp
f3	¬q	mt(f1, f2)
f4	q ∨ t	hyp
f5	t	ds(f4, f3)
f6	t → r	hyp
f7	r	mp(f5, f6)

10.4 p ∨ (q → r), p ∨ q, ¬p ⊢ r

	Formula	Justification
f1	p ∨ (q → r)	hyp
f2	¬p	hyp
f3	q → r	ds(f1, f2)
f4	p ∨ q	hyp
f5	q	ds(f4, f2)
f6	r	mp(f5, f3)

10.5 ¬p ∨ ¬q ∨ r, q, p ⊢ r

	Formula	Justification
f1	¬p ∨ (¬q ∨ r)	hyp (associativity of ∨)
f2	p	hyp
f3	¬¬p	f2, U ≡ ¬¬U
f4	¬q ∨ r	ds(f1, f3)
f5	q	hyp
f6	¬¬q	f5
f7	r	ds(f4, f6)

10.6 p → ¬q ∨ r, p ∧ q, p ⊢ r

	Formula	Justification
f1	p	hyp
f2	p → ¬q ∨ r	hyp
f3	¬q ∨ r	mp(f1, f2)
f4	p ∧ q	hyp
f5	q	∧e(f4)
f6	¬¬q	f5
f7	r	ds(f3, f6)

14. Exercise 11 — Is C deducible? (natural language)

In all three sets C is deducible. Method: (1) choose variables, (2) formalize the hypotheses and C, (3) build the deduction.

Set 1 — swimming / canoe

s: "it is sunny this afternoon", c: "it is colder than yesterday", w: "we go swimming", k: "we take a canoe trip", h: "we will be home by sunset".

H1 = ¬s ∧ c · H2 = w → s · H3 = ¬w → k · H4 = k → h · C = h.

	Formula	Justification
f1	¬s ∧ c	H1
f2	¬s	∧e(f1)
f3	w → s	H2
f4	¬w	mt(f3, f2)
f5	¬w → k	H3
f6	k	mp(f4, f5)
f7	k → h	H4
f8	h	mp(f6, f7)

Conclusion: H1, H2, H3, H4 ⊢ C — yes, we will be home by sunset.

Set 2 — Alice and the bank

r: "Alice robbed the bank", l: "Alice left the house last night", i: "the police investigate the employees", b: "the police question Bob".

H1 = r → l · H2 = ¬r → i · H3 = i → b · H4 = ¬l · C = b.

	Formula	Justification
f1	r → l	H1
f2	¬l	H4
f3	¬r	mt(f1, f2)
f4	¬r → i	H2

	Formula	Justification
f5	i	mp(f3, f4)
f6	i → b	H3
f7	b	mp(f5, f6)

Conclusion: C is deducible — the police will question Bob.

Set 3 — the dragon

c: "the dragon lives for many centuries", t: "he accumulates a vast treasure", u: "his lair remains undisturbed", w: "the neighbouring kingdoms are at war".

H1 = c → t · H2 = t → u ("only if") · H3 = w → ¬u ("provided that" = if) · H4 = c · C = ¬w.

	Formula	Justification
f1	c	H4
f2	c → t	H1
f3	t	mp(f1, f2)
f4	t → u	H2
f5	u	mp(f3, f4)
f6	¬¬u	f5, U ≡ ¬¬U
f7	w → ¬u	H3
f8	¬w	mt(f7, f6)

Conclusion: C is deducible — the kingdoms are not at war.

15. Exercise 12 — Theorems via the deduction theorem

Each proof: hypotheses → deduction of the last consequent → apply the deduction theorem (DT) once per premise moved.

12.1 ⊢ p ∨ (q → r) → ((p ∨ q) → (p ∨ r))

Rewrite with X ∨ Y ≡ ¬X → Y: p ∨ (q → r) ≡ ¬p → (q → r); p ∨ q ≡ ¬p → q; p ∨ r ≡ ¬p → r. It suffices to prove ¬p → (q → r), ¬p → q, ¬p ⊢ r:

	Formula	Justification
f1	¬p	hyp
f2	¬p → (q → r)	hyp
f3	q → r	mp(f1, f2)
f4	¬p → q	hyp
f5	q	mp(f1, f4)
f6	r	mp(f5, f3)

DT: ¬p → (q → r), ¬p → q ⊢ ¬p → r, i.e. p ∨ (q → r), p ∨ q ⊢ p ∨ r. DT twice more → the theorem.

12.2 ⊢ (p → (¬r → q)) → (r ∨ ¬p ∨ q)

r ∨ (¬p ∨ q) ≡ ¬r → (¬p ∨ q) ≡ ¬r → (p → q). Prove p → (¬r → q), ¬r, p ⊢ q:

	Formula	Justification
f1	p	hyp
f2	p → (¬r → q)	hyp
f3	¬r → q	mp(f1, f2)
f4	¬r	hyp
f5	q	mp(f4, f3)

DT (remove p): …, ¬r ⊢ p → q; DT (remove ¬r): p → (¬r → q) ⊢ ¬r → (p → q) ≡ r ∨ ¬p ∨ q; DT once more → the theorem.

12.3 ⊢ (p → (q → r)) → (p ∧ q → r) — prove p → (q → r), p ∧ q ⊢ r: f1 p ∧ q (hyp); f2 p, f3 q (∧e); f4 p → (q → r) (hyp); f5 q → r mp(f2, f4); f6 r mp(f3, f5). DT twice.

12.4 ⊢ (p ∧ q → r) → (p → (q → r)) — prove p ∧ q → r, p, q ⊢ r: f1 p, f2 q (hyp); f3 p ∧ q ∧i(f1, f2); f4 p ∧ q → r (hyp); f5 r mp(f3, f4). DT three times (remove q, then p, then p ∧ q → r).

12.5 ⊢ (p → (q → r)) → (q → (p → r)) — prove p → (q → r), q, p ⊢ r: f1 p (hyp); f2 p → (q → r) (hyp); f3 q → r mp(f1, f2); f4 q (hyp); f5 r mp(f4, f3). DT removing p, then q, then the first hypothesis.

12.6 ⊢ (p → (q → r)) → ((p → q) → (p → r)) — prove p → (q → r), p → q, p ⊢ r: f1 p (hyp); f2 p → (q → r) (hyp); f3 q → r mp(f1, f2); f4 p → q (hyp); f5 q mp(f1, f4); f6 r mp(f5, f3). DT three times. (This is axiom A2 with U = p, V = q, Z = r — you may also simply cite A2.)

16. Exercise 13 — More theorems via the deduction theorem

13.1 ⊢ (p → (q ∨ r)) → ((p → q) ∨ (p → r))

(p → q) ∨ (p → r) ≡ ¬(p → q) → (p → r), and ¬(p → q) ≡ p ∧ ¬q. Prove p → q ∨ r, ¬(p → q), p ⊢ r:

	Formula	Justification
f1	p	hyp
f2	p → q ∨ r	hyp
f3	q ∨ r	mp(f1, f2)
f4	¬(p → q)	hyp
f5	p ∧ ¬q	f4, equivalence
f6	¬q	∧e(f5)
f7	r	ds(f3, f6)

DT removing p, then ¬(p → q), then p → q ∨ r → the theorem.

13.2 ⊢ (p → q) → ((¬r ∨ p) → (r → q)) — prove p → q, ¬r ∨ p, r ⊢ q: f1 r (hyp); f2 ¬¬r (f1); f3 ¬r ∨ p (hyp); f4 p ds(f3, f2); f5 p → q (hyp); f6 q mp(f4, f5). DT three times.

13.3 ⊢ p ∨ (q → r) → ((p ∨ q) → (p ∨ r)) — identical to 12.1.

13.4 ⊢ (p → r) → ((q → r) → (p ∨ q → r)) — prove p → r, q → r, p ∨ q ⊢ r:

	Formula	Justification
f1	p ∨ q	hyp
f2	¬p → q	f1, U ∨ V ≡ ¬U → V
f3	q → r	hyp
f4	¬p → r	sh(f2, f3)
f5	p → r	hyp
f6	(p → r) → (¬r → ¬p)	A3
f7	¬r → ¬p	mp(f5, f6)
f8	¬r → r	sh(f7, f4)
f9	r ∨ r	f8, ¬U → V ≡ U ∨ V
f10	r	f9, idempotence

DT three times → the theorem.

13.5 ⊢ (p → q) → ((r → t) → (p ∧ r → q ∧ t)) — prove p → q, r → t, p ∧ r ⊢ q ∧ t: f1 p ∧ r (hyp); f2 p, f3 r (∧e); f4 q mp(f2, p → q); f5 t mp(f3, r → t); f6 q ∧ t ∧i(f4, f5). DT three times.

13.6 ⊢ (p → r) → ((p ∧ r → q) → (p → q)) — prove p → r, p ∧ r → q, p ⊢ q: f1 p (hyp); f2 r mp(f1, p → r); f3 p ∧ r ∧i(f1, f2); f4 q mp(f3, p ∧ r → q). DT three times.

17. Theory — Predicate (first-order) logic

Predicate logic adds objects, properties and quantifiers to propositional logic; truth is now evaluated in an interpretation over a domain, not in a truth-table row.

17.1 Syntax

- **Variables:** x, y, z, … range over the domain D.
- **Constants:** a, b, 1, John, … name fixed elements of D.
- **Function symbols** (arity n ⩾ 1): sum(x, y), prod(x, y), sq(x), father(x) — return an **element** of D.
- **Predicate symbols** (arity n ⩾ 1): even(x), eq(x, y), Likes(x, y) — return a **truth value**.
- **Terms:** variables, constants, and $f(t_1, \ldots, t_n)$. **Atoms:** $P(t_1, \ldots, t_n)$. Formulas are built from atoms with ¬ ∧ ∨ → ↔ and the quantifiers (∀x) "for all", (∃x) "there exists".
- **Free / bound variables:** an occurrence of x is bound if it is inside the scope of (∀x) or (∃x); a formula with no free variables is a **closed formula (sentence)**.

17.2 Formalization patterns

Natural language	Formula
All A are B	(∀x)(A(x) → B(x))
Some A are B	(∃x)(A(x) ∧ B(x))
No A is B	(∀x)(A(x) → ¬B(x)) ≡ ¬(∃x)(A(x) ∧ B(x))
Only A are B	(∀x)(B(x) → A(x))
Not all A are B	(∃x)(A(x) ∧ ¬B(x))

Classic trap: ∀ goes with →, ∃ goes with ∧. (∀x)(A(x) ∧ B(x)) says *everything* is both A and B; (∃x)(A(x) → B(x)) is true as soon as one element is not A.

17.3 Semantics

- **Interpretation** I = ⟨D, m⟩: a non-empty domain D and a mapping m giving each constant an element m(a) ∈ D, each n-ary function symbol a function m(f): D^n → D, each n-ary predicate symbol a function m(P): D^n → {T, F}.
- **Evaluation** v^I: atoms by m; connectives as in propositional logic; v^I((∀x)U) = T iff U is T for **every** d ∈ D; v^I((∃x)U) = T iff U is T for **at least one** d ∈ D.
- **Witness:** to prove (∀x)U false, or (∃x)U true, give one element d. To prove (∀x)U true, or (∃x)U false, argue for an arbitrary element.
- On a finite domain D = {a, b}: (∀x)A(x) ≡ A(a) ∧ A(b) and (∃x)A(x) ≡ A(a) ∨ A(b). This turns a predicate formula into a propositional one — the main tool in Ex 5–6.
- **Model / anti-model** of U: an interpretation I with v^I(U) = T / F. **Valid:** every interpretation is a model. **Consistent:** at least one model. **Inconsistent:** none. **Contingent:** has both.

17.4 Inference rules for quantifiers (Ex 3)

- **Universal instantiation (∀e):** (∀x)U(x) ⊢ U(t) for any term t (e.g. a constant).
- **Existential generalization (∃i):** U(t) ⊢ (∃x)U(x).
- Combine with mp, ∧i, ∧e from §12.2.

18. Predicate Exercise 1 — Mathematical statements

One formula per sentence, all over the domain D = ℤ (ℝ for 1.3). Shared vocabulary, declared once as the sheet requires:

Kind	Symbols	Meaning
Variables	x, y, z, u, w	elements of D
Constants	0, 1, 2, 3, 4, 5, 6, 10	the numbers
Function symbols (arity 2)	sum(x, y), diff(x, y), prod(x, y)	x + y, x − y, x · y
Predicate symbols (arity 1)	even(x), pos(x), neg(x), prime(x)	x even, x > 0, x < 0, x prime
Predicate symbols (arity 2)	eq(x, y), div(x, y), less(x, y)	x = y, x divisible by y, x < y

Function symbols build numbers, predicate symbols build statements: div(prod(x, y), 4) is well-formed; even(prod(x, y)) is well-formed; prod(even(x), y) is not.

1.1 The sum and difference of two even numbers is even and their product is divisible by 4.

(∀x)(∀y)(even(x) ∧ even(y) → even(sum(x, y)) ∧ even(diff(x, y)) ∧ div(prod(x, y), 4))

1.2 If x is a nonzero integer divisible by 10, it splits into two factors, one divisible by 2 and the other by 5, and x is a sum of 2 even numbers.

(∀x)(¬eq(x, 0) ∧ div(x, 10) → (∃y)(∃z)(eq(x, prod(y, z)) ∧ div(y, 2) ∧ div(z, 5)) ∧ (∃u)(∃w)(even(u) ∧ even(w) ∧ eq(x, sum(u, w))))

1.3 Every positive number is a product of two positive numbers and a product of two negative numbers (D = ℝ).

(∀x)(pos(x) → (∃y)(∃z)(pos(y) ∧ pos(z) ∧ eq(x, prod(y, z))) ∧ (∃u)(∃w)(neg(u) ∧ neg(w) ∧ eq(x, prod(u, w))))

1.4 If x and y are positive primes, x ≠ y, x ≠ 2, y ≠ 2, their sum and difference are even and their product is odd.

(∀x)(∀y)(pos(x) ∧ pos(y) ∧ prime(x) ∧ prime(y) ∧ ¬eq(x, y) ∧ ¬eq(x, 2) ∧ ¬eq(y, 2) → even(sum(x, y)) ∧ even(diff(x, y)) ∧ ¬even(prod(x, y)))

1.5 (seminar) For every positive integer x, if x is a square of an integer, then there is an integer y with (y + 1)(y − 1) = x − 1.

(∀x)(pos(x) ∧ (∃z)eq(x, prod(z, z)) → (∃y)eq(prod(sum(y, 1), diff(y, 1)), diff(x, 1)))

The seminar wrote addition/substraction/product — any names work if you declare them. The seminar also listed properties of the functions, e.g. commutativity (∀x)(∀y)eq(sum(x, y), sum(y, x)), associativity (∀x)(∀y)(∀z)eq(sum(sum(x, y), z), sum(x, sum(y, z))), identity element (∀x)eq(sum(x, 0), x), distributivity (∀x)(∀y)(∀z)eq(prod(x, sum(y, z)), sum(prod(x, y), prod(x, z))).

1.6 If x is divisible by 3 and y by 6, their product is even and their sum is divisible by 3.

(∀x)(∀y)(div(x, 3) ∧ div(y, 6) → even(prod(x, y)) ∧ div(sum(x, y), 3))

1.7 For every positive integer x that is not prime, there is a prime y that divides x with y < x.

(∀x)(pos(x) ∧ ¬prime(x) → (∃y)(prime(y) ∧ div(x, y) ∧ less(y, x)))

19. Predicate Exercise 2 — Everyday statements

Predicates are named after the property; constants are proper names (John, Bill, Santa, Algebra …). Domain: all people, animals and objects mentioned. Pattern reminder: ∀ with →, ∃ with ∧ (§17.2).

#	Statement	Formula
1	Every student who makes good grades is brilliant or studies.	(∀x)(Student(x) ∧ Grades(x) → Brilliant(x) ∨ Studies(x))
2	Some of John's colleagues like to draw and some like to dance.	(∃x)(Colleague(x, John) ∧ Draw(x)) ∧ (∃y)(Colleague(y, John) ∧ Dance(y))
3	CS students like either algebra or logic, all like Java, but only Bill likes history.	(∀x)(CS(x) → Likes(x, Algebra) ∨ Likes(x, Logic)) ∧ (∀x)(CS(x) → Likes(x, Java)) ∧ CS(Bill) ∧ Likes(Bill, History) ∧ (∀x)(CS(x) ∧ Likes(x, History) → eq(x, Bill))
4	All Mary's relatives live in Cluj-Napoca, only her cousin John lives in Bucharest.	Cousin(John, Mary) ∧ LivesIn(John, Bucharest) ∧ (∀x)(Relative(x, Mary) ∧ ¬eq(x, John) → LivesIn(x, Cluj))
5	Caterpillars and snails like to eat some plants.	(∀x)(Caterpillar(x) ∨ Snail(x) → (∃y)(Plant(y) ∧ Eats(x, y)))
6	Anyone who owns a rabbit hates anything that chases a rabbit.	(∀x)((∃y)(Rabbit(y) ∧ Owns(x, y)) → (∀z)((∃w)(Rabbit(w) ∧ Chases(z, w)) → Hates(x, z)))
7	Anyone who loves some candy is not a nutrition fanatic.	(∀x)((∃y)(Candy(y) ∧ Loves(x, y)) → ¬Fanatic(x))
8	If Santa has some reindeer with a red nose, every child loves Santa.	(∃x)(Reindeer(x) ∧ Has(Santa, x) ∧ RedNose(x)) → (∀y)(Child(y) → Loves(y, Santa))
9	Anyone who has any cats will not have any mice.	(∀x)((∃y)(Cat(y) ∧ Has(x, y)) → ¬(∃z)(Mouse(z) ∧ Has(x, z)))

#	Statement	Formula
10	Caterpillars and snails ≪ birds ≪ foxes ≪ wolves.	(∀x)(∀y)((Caterpillar(x) ∨ Snail(x)) ∧ Bird(y) → Smaller(x, y)) ∧ (∀x)(∀y)(Bird(x) ∧ Fox(y) → Smaller(x, y)) ∧ (∀x)(∀y)(Fox(x) ∧ Wolf(y) → Smaller(x, y))
11	Every animal likes to eat all plants or all much-smaller animals that eat some plants.	(∀x)(Animal(x) → (∀y)(Plant(y) → Eats(x, y)) ∨ (∀y)(Animal(y) ∧ Smaller(y, x) ∧ (∃z)(Plant(z) ∧ Eats(y, z)) → Eats(x, y)))
12	If every roommate of every CS major likes to party, every CS-major student with good grades is brilliant.	(∀x)(∀y)(CS(x) ∧ Roommate(y, x) → Party(y)) → (∀z)(Student(z) ∧ CS(z) ∧ Grades(z) → Brilliant(z))
13	If Santa is happy at Christmas, every good child gets some toy at Christmas.	Happy(Santa, Christmas) → (∀x)(Child(x) ∧ Good(x) → (∃y)(Toy(y) ∧ Gets(x, y, Christmas)))
14	Wolves don't eat foxes or grains; birds eat caterpillars but not snails.	(∀x)(∀y)(Wolf(x) ∧ (Fox(y) ∨ Grain(y)) → ¬Eats(x, y)) ∧ (∀x)(∀y)(Bird(x) ∧ Caterpillar(y) → Eats(x, y)) ∧ (∀x)(∀y)(Bird(x) ∧ Snail(y) → ¬Eats(x, y))

Why rows 6, 7, 9 use ∃ inside the premise: "anyone who owns **a** rabbit" means "owns at least one". (∀x)((∃y)(Rabbit(y) ∧ Owns(x, y)) → …) ≡ (∀x)(∀y)(Rabbit(y) ∧ Owns(x, y) → …) — both forms are correct.

20. Predicate Exercise 3 — Succession to the British throne

Yes — HenryVIII can become the king. Vocabulary: constants R = RichardIII, H7 = HenryVII, H8 = HenryVIII; King(x) "x is (becomes) the king", Son(y, x) "y is the oldest son of x", Defeats(y, x), CanBeKing(y).

- H1 = (∀x)(∀y)(King(x) ∧ Son(y, x) → CanBeKing(y))
- H2 = (∀x)(∀y)(King(x) ∧ Defeats(y, x) → King(y))
- H3 = King(R) · H4 = Defeats(H7, R) · H5 = Son(H8, H7) · C = CanBeKing(H8)

	Formula	Justification
f1	(∀x)(∀y)(King(x) ∧ Defeats(y, x) → King(y))	H2
f2	King(R) ∧ Defeats(H7, R) → King(H7)	∀e(f1), x := R, y := H7
f3	King(R)	H3

	Formula	Justification
f4	Defeats(H7, R)	H4
f5	King(R) ∧ Defeats(H7, R)	∧i(f3, f4)
f6	King(H7)	mp(f5, f2)
f7	(∀x)(∀y)(King(x) ∧ Son(y, x) → CanBeKing(y))	H1
f8	King(H7) ∧ Son(H8, H7) → CanBeKing(H8)	∀e(f7), x := H7, y := H8
f9	Son(H8, H7)	H5
f10	King(H7) ∧ Son(H8, H7)	∧i(f6, f9)
f11	CanBeKing(H8)	mp(f10, f8)

Conclusion: H1, …, H5 ⊢ C. The key step is H2 instantiated first: it makes HenryVII king, which H1 needs.

21. Predicate Exercise 4 — Evaluate under a given interpretation

Method (§17.3): split U at its main connective, evaluate each quantified subformula on the domain — one witness decides a false ∀ or a true ∃ — then combine.

	Formula	Value	Decisive step
(a)	(∀x)(∀y)(A(x) ∧ A(y) → ¬A(sum(x, y)) ∧ (∀z)A(prod(y, z)))	**F**	x = y = 1, z = 2: 1·2 = 2 is not odd
(b)	(∃x)(P(x) ∧ Q(x)) → (∃x)P(x) ∨ Q(12)	**T**	consequent true: 5 is divisible by 5
(c)	(∃x)A(x) ∧ (∃x)B(x) → (∀x)(A(x) ∨ B(x))	**F**	a line at 45° to d is neither ⊥ nor ∥
(d)	(∀x)(P(x) ∧ Q(x) → P(sq(x)) ∧ Q(prod(x, 5)))	**T**	x even, x < 0 ⇒ x^2 even, 5x < 0
(e)	(∀x)(A(x) → B(x)) ∧ (∃x)(B(x) ∧ ¬A(x))	**F**	x = 2 is prime but not odd
(f)	((∀x)A(x) → (∃x)B(x)) → ((∃x)A(x) → (∀x)B(x))	**F**	antecedent T, consequent T → F
(g)	(∃x)A(x) ∧ (∃x)B(x) → (∀x)(A(x) ∧ B(x))	**F**	not every person owns a car

(a) D = ℕ, A(x): "x is odd", sum = +, prod = ·.

- v^I(U) = (∀x)(∀y)("x odd" ∧ "y odd" → "x + y not odd" ∧ (∀z)"y·z odd").
- Take x = 1, y = 1: premise T ∧ T = T. First conjunct: 1 + 1 = 2 is not odd → T. Second: for z = 2, 1·2 = 2 is not odd → (∀z)A(prod(1, z)) = F.
- Consequent T ∧ F = F, so the implication is T → F = F for x = y = 1.
- **Conclusion:** v^I(U) = F, I is an anti-model of U.

(b) D = ℕ, P(x): "x ⋮ 5", Q(x): "x ⋮ 7".

- (∃x)(P(x) ∧ Q(x)): x = 35 is divisible by 5 and 7 → T.
- (∃x)P(x): x = 5 → T, so (∃x)P(x) ∨ Q(12) = T ∨ F = T.
- **Conclusion:** v^I(U) = T → T = T, I is a model.

(c) D = the lines of plane P, d fixed, A(x): "x ⊥ d", B(x): "x ∥ d".

- (∃x)A(x) = T (any perpendicular to d); (∃x)B(x) = T (any parallel to d).
- (∀x)(A(x) ∨ B(x)) = F: a line forming a 45° angle with d is neither perpendicular nor parallel.
- **Conclusion:** v^I(U) = T ∧ T → F = F, anti-model.

(d) (seminar) D = ℤ, P(x): "x even", Q(x): "x < 0", sq(x) = x², prod = ·.

- v^I(U) = (∀x)("x even" ∧ "x < 0" → "x² even" ∧ "5x < 0").
- If x is even, x = 2k, so x² = 4k² is even. If x < 0, then 5x < 0. So whenever the premise is T the consequent is T; when the premise is F the implication is T.
- **Conclusion:** v^I(U) = T, model.

(e) (seminar) D = ℕ, A(x): "x prime", B(x): "x odd".

- U_1 = (∀x)(A(x) → B(x)): x = 2 is prime and not odd → T → F = F, so U_1 = F.
- U_2 = (∃x)(B(x) ∧ ¬A(x)): x = 9 is odd and not prime → T.
- **Conclusion:** v^I(U) = U_1 ∧ U_2 = F ∧ T = F, anti-model. *(The seminar notes stop at U_2 = T — don't forget the final ∧.)*

(f) D = persons from Romania, A(x): "x likes chess", B(x): "x is a student".

- (∀x)A(x) = F (not everybody likes chess) → (∀x)A(x) → (∃x)B(x) = T.
- (∃x)A(x) = T (some people like chess); (∀x)B(x) = F (not everybody is a student) → (∃x)A(x) → (∀x)B(x) = F.
- **Conclusion:** v^I(U) = T → F = F, anti-model.

(g) D = persons from Romania, A(x): "x owns a car", B(x): "x has a driver licence".

- (∃x)A(x) = T, (∃x)B(x) = T; (∀x)(A(x) ∧ B(x)) = F (e.g. a child has neither).
- **Conclusion:** v^I(U) = T ∧ T → F = F, anti-model.

22. Predicate Exercise 5 — A model on a 2-element domain

One interpretation is a model of all eight formulas: I = ⟨D, m⟩, D = {1, 2}, m(A)(x): "x = 1", m(B)(x): "x > 0". In fact all eight are valid, so any interpretation would work — but the exercise asks for one, verified.

On a 2-element domain replace each quantifier by its expansion (§17.3): (∀x)A(x) = A(1) ∧ A(2), (∃x)A(x) = A(1) ∨ A(2). Values: A(1) = T, A(2) = F, B(1) = T, B(2) = T.

Subformula	Expansion	Value
(∀x)A(x)	T ∧ F	F
(∃x)A(x)	T ∨ F	T
(∀x)B(x), (∃x)B(x)	T ∧ T, T ∨ T	T, T
(∀x)(A(x) ↔ B(x))	(T ↔ T) ∧ (F ↔ T)	F
(∀x)(A(x) → B(x))	(T → T) ∧ (F → T)	T
(∃x)(A(x) → B(x))	(T → T) ∨ (F → T)	T
(∀x)(A(x) ∨ B(x))	(T ∨ T) ∧ (F ∨ T)	T

Formula	Evaluation	v^I
U_1 = (∀x)(A ↔ B) → ((∀x)A ↔ (∀x)B)	F → (F ↔ T)	T
U_2 = (∀x)(A → B) → ((∀x)A → (∀x)B)	T → (F → T) = T → T	T
U_3 = (∀x)(A ↔ B) → ((∃x)A ↔ (∃x)B)	F → (T ↔ T)	T
U_4 = (∃x)(A → B) ↔ ((∀x)A → (∃x)B)	T ↔ (F → T) = T ↔ T	T
U_5 = ((∃x)A → (∀x)B) → (∀x)(A → B)	(T → T) → T = T → T	T
U_6 = (∀x)(A ∨ B) → ((∀x)A ∨ (∃x)B)	T → (F ∨ T) = T → T	T
U_7 = (∀x)(A → B) → ((∃x)A → (∃x)B)	T → (T → T) = T → T	T
U_8 = (∀x)(A → B) → ((∀x)A → (∃x)B)	T → (F → T) = T → T	T

Conclusion: I is a model of U_1 … U_8. U_1 and U_3 are true only because their antecedent is false; if you want a non-trivial check, the seminar's interpretation for U_5 (A(x): "x < 3", B(x): "x > −1", so everything is T) works as well.

23. Predicate Exercise 6 — A model and an anti-model

Strategy (seminar): for U = A → B, an anti-model needs v(A) = T and v(B) = F; a model needs v(A) = F or v(B) = T. Domain D = {1, 2} throughout; predicates chosen from "x > 0" (T, T), "x = 1" (T, F), "x = 2" (F, T), "x < 0" (F, F).

Formula	Model: P(x), Q(x)	Anti-model: P(x), Q(x)
$U_1 = ((\exists x)P \to (\exists x)Q) \to (\forall x)(P \to Q)$	x > 0, x > 0	x > 0, x = 1
$U_2 = (\exists x)(P \to Q) \to ((\exists x)P \to (\exists x)Q)$	x > 0, x > 0	x = 1, x < 0
$U_3 = ((\forall x)P \to (\forall x)Q) \to (\forall x)(P \to Q)$	x > 0, x > 0	x = 1, x = 2
$U_4 = (\exists x)P \wedge (\exists x)Q \to (\exists x)(P \wedge Q)$	x > 0, x > 0	x = 1, x = 2
$U_5 = (\forall x)(P \vee Q) \to (\forall x)P \vee (\forall x)Q$	x > 0, x < 0	x = 1, x = 2
$U_6 = ((\forall x)P \to (\exists x)Q) \to (\forall x)(P \to Q)$	x > 0, x > 0	x > 0, x = 1
$U_7 = (\exists x)P \wedge (\exists x)Q \to (\forall x)(P \wedge Q)$	x > 0, x > 0	x = 1, x = 1

With P = Q = "x > 0" every atom is T, so every consequent above is T and every formula is T — one model for all seven. The anti-models need checking one by one:

- **U_1** (P: x > 0, Q: x = 1): $(\exists x)P \to (\exists x)Q = T \to T = T$; $(\forall x)(P \to Q) = (T \to T) \wedge (T \to F) = F$. v = T → F = **F**.
- **U_2** (P: x = 1, Q: x < 0): $(\exists x)(P \to Q) = (T \to F) \vee (F \to F) = T$; $(\exists x)P \to (\exists x)Q = T \to F = F$. v = T → F = **F**.
- **U_3** (P: x = 1, Q: x = 2): $(\forall x)P \to (\forall x)Q = F \to F = T$; $(\forall x)(P \to Q) = (T \to F) \wedge \ldots = F$. v = **F**.
- **U_4** (P: x = 1, Q: x = 2): $(\exists x)P \wedge (\exists x)Q = T \wedge T = T$; $(\exists x)(P \wedge Q) = (T \wedge F) \vee (F \wedge T) = F$. v = **F**.
- **U_5** (seminar; P: x = 1, Q: x = 2): $(\forall x)(P \vee Q) = (T \vee F) \wedge (F \vee T) = T$; $(\forall x)P \vee (\forall x)Q = (T \wedge F) \vee (F \wedge T) = F$. v = **F**. Model (seminar): P: x > 0, Q: x < 0 → T → (T ∨ F) = T.
- **U_6** (P: x > 0, Q: x = 1): $(\forall x)P \to (\exists x)Q = T \to T = T$; $(\forall x)(P \to Q) = (T \to T) \wedge (T \to F) = F$. v = **F**.
- **U_7** (P: x = 1, Q: x = 1): $(\exists x)P \wedge (\exists x)Q = T$; $(\forall x)(P \wedge Q) = T \wedge F = F$. v = **F**.

Conclusion: all seven formulas are contingent — each has a model and an anti-model, so none is valid.

24. Semantic tableaux — theory and the seminar example

A tableau searches for models by breaking a formula into literals along branches: a branch is **closed** (⊗) if it contains a pair p, ¬p, and **open** (⊙) otherwise; every open branch gives models.

24.1 Rules

α-rules (conjunctive: write both, one under the other)	β-rules (disjunctive: split into two branches)
A ∧ B → A, B	A ∨ B → A | B
¬(A ∨ B) → ¬A, ¬B	¬(A ∧ B) → ¬A | ¬B
¬(A → B) → A, ¬B	A → B → ¬A | B
¬¬A → A	A ↔ B → (A ∧ B) | (¬A ∧ ¬B)
A ↔ B → A → B, B → A	¬(A ↔ B) → (A ∧ ¬B) | (¬A ∧ B)

Tip: apply α-rules before β-rules — fewer branches to carry.

24.2 What the tableau tells you

Tableau of U	Conclusion
all branches closed	U is **inconsistent**
at least one open branch	U is **consistent**; each open branch's literals form a cube of a DNF of U → read the models
tableau of **¬U** closes completely	U is **valid** (to prove validity, build the tableau of the negation)

24.3 Seminar example

U = (r ∨ q) ∨ (p → ¬r) → (p ↔ q). Decide what kind of formula it is; if consistent, find its models.

1. β-rule on → (1): left branch ¬[(r ∨ q) ∨ (p → ¬r)] (2); right branch p ↔ q (3).
2. Left: α on (2) gives ¬(r ∨ q) (4) and ¬(p → ¬r) (5); α on (4) gives ¬r, ¬q; α on (5) gives p, r. The branch holds r and ¬r → **closed** ⊗.
3. Right: p ↔ q ≡ (p → q) ∧ (q → p); α gives p → q (6) and q → p (7). β on (6): ¬p | q. β on (7) under each:
   - ¬p, ¬q → **open** ⊙
   - ¬p, p → closed ⊗
   - q, ¬q → closed ⊗
   - q, p → **open** ⊙
4. Open branches → DNF(U) = (¬p ∧ ¬q) ∨ (p ∧ q).

5. Models: ¬p ∧ ¬q → i7, i8 (r free); p ∧ q → i1, i2. E.g. i7(p) = F, i7(q) = F, i7(r) = T.

Conclusion: U is **consistent** (open branches exist) and **not valid** — e.g. i3 (p = T, q = F) is an anti-model. Its models are i1, i2, i7, i8; U is contingent. (Checked against the truth table.)

24.4 Practice suggestion

Re-do Exercise 1 and Exercise 8 with tableaux: in Ex 1 the open branches must give exactly the models listed in §2; in Ex 8 every branch must close.